

RJEŠENJA ISPITA DRŽAVNE MATURE IZ **KEMIJE**
U ŠKOLSKOJ GODINI 2020./2021. (jesenski rok)

D-S047

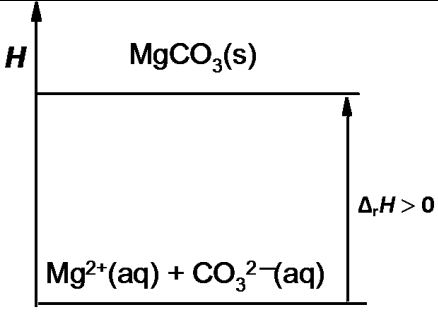
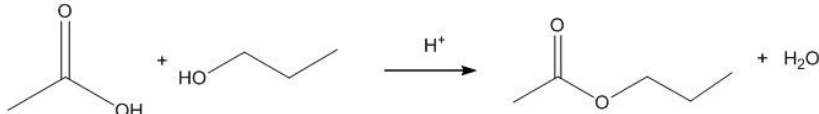
ISPITNA KNJIŽICA 1

BROJ ZADATKA	TOČAN ODGOVOR
1.	C
2.	C
3.	D
4.	D
5.	D
6.	C
7.	C
8.	A
9.	A
10.	D
11.	B
12.	B
13.	A
14.	C
15.	A
16.	C
17.	D
18.	D
19.	C
20.	A
21.	C
22.	A
23.	A
24.	C
25.	C
26.	C
27.	C
28.	C
29.	C
30.	D
31.	B
32.	C
33.	B
34.	A
35.	A

ISPITNA KNJIŽICA 2

BROJ ZADATKA	TOČAN ODGOVOR	BOD
1.1.	but-2-en	1 BOD
1.2.	Li ₂ O	1 BOD
2.1.	volumno-centriranoj	1 BOD
2.2.	$\rho = 1,53 \text{ g cm}^{-3}$ $\rho = \frac{N \cdot Ar(Rb) \cdot u}{a^3} = \frac{2 \cdot 85,5 \cdot 1,66 \times 10^{-24} \text{ g}}{(5,7 \times 10^{-8} \text{ cm})^3} = 1,53 \text{ g cm}^{-3}$	1 BOD
3.1.	I, (+1)	1 BOD
3.2.	$\text{Cl}_2 + 2 \text{H}_2\text{O} \rightarrow 2 \text{ClO}^- + 4 \text{H}^+ + 2 \text{e}^-$	1 BOD
4.	$c_0 = 0,11 \text{ mol dm}^{-3}$ $\alpha = 4,1 \% (4,0 \pm 0,1 \%)$ $[\text{H}_3\text{O}^+] = 10^{-\text{pH}} \text{ mol dm}^{-3} = 10^{-2,35} \text{ mol dm}^{-3} = 4,47 \times 10^{-3} \text{ mol dm}^{-3}$ $K_a = \frac{[\text{H}_3\text{O}^+] \cdot [\text{HCOO}^-]}{[\text{HCOOH}]}$ $c_0 = \frac{[\text{H}_3\text{O}^+]^2}{K_a} = \frac{(4,47 \times 10^{-3} \text{ mol dm}^{-3})^2}{1,8 \times 10^{-4} \text{ mol dm}^{-3}} = 0,11 \text{ mol dm}^{-3}$ $\alpha = \frac{[\text{H}_3\text{O}^+]}{c_0} = \frac{4,47 \times 10^{-3} \text{ mol dm}^{-3}}{0,11 \text{ mol dm}^{-3}} \cdot 100 = 4,1 \%$	1 BOD 1 BOD
5.1.	$m_f(\text{C}_{14}\text{H}_{10}) = 2,96 \times 10^{-25} \text{ kg}$ $M_r(\text{C}_{14}\text{H}_{10}) = \frac{m_f(\text{C}_{14}\text{H}_{10})}{u} \Rightarrow m_f(\text{C}_{14}\text{H}_{10}) = M_r(\text{C}_{14}\text{H}_{10}) \cdot u$ $m_f(\text{C}_{14}\text{H}_{10}) = 178,1 \cdot 1,66 \times 10^{-27} \text{ kg} = 2,96 \times 10^{-25} \text{ kg}$	1 BOD
5.2.	$w(\text{H}, \text{C}_{14}\text{H}_{10}) = 5,67 \%$	1 BOD

	$w(\text{H}, \text{C}_{14}\text{H}_{10}) = \frac{10 A_r(\text{H})}{M_r(\text{C}_{14}\text{H}_{10})} = \frac{10 \cdot 1,01}{178,1} = 0,0567 \cdot 100 \% = 5,67 \%$	
6.	$m(\text{NH}_3) = 251,44 \text{ g}$ $n(\text{NH}_3) = 14,76 \text{ mol}$ $V(\text{NH}_3) = 0,36 \text{ m}^3 (0,35 \pm 0,01 \text{ m}^3)$ $m(\text{NH}_3) = w(\text{NH}_3) \cdot m(\text{otopina}) = w(\text{NH}_3) \cdot \rho(\text{otopina}) \cdot V(\text{otopina})$ $m(\text{NH}_3) = 0,28 \cdot 0,898 \text{ g cm}^{-3} \cdot 1000 \text{ cm}^3 = 251,44 \text{ g}$ $n(\text{NH}_3) = \frac{m(\text{NH}_3)}{M(\text{NH}_3)} = \frac{251,44 \text{ g}}{17,03 \text{ g mol}^{-1}} = 14,76 \text{ mol}$ $p \cdot V = n \cdot R \cdot T$ $V = \frac{n \cdot R \cdot T}{p}$ $V = \frac{14,76 \text{ mol} \cdot 8,31 \text{ Pa m}^3 \text{ K}^{-1} \text{ mol}^{-1} \cdot 293 \text{ K}}{101\,300 \text{ Pa}} = 0,36 \text{ m}^3$	1 BOD 1 BOD 1 BOD
7.1.	$V(\text{HCl}) = 40 \text{ mL}$	1 BOD
7.2.	žute	1 BOD
7.3.	$c(\text{NaOH}) = 0,0910 \text{ mol dm}^{-3}$ $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O}(\ell)$ $n(\text{HCl}) = n(\text{NaOH})$ $c(\text{HCl}) \cdot V(\text{HCl}) = c(\text{NaOH}) \cdot V(\text{NaOH})$ $c(\text{NaOH}) = \frac{c(\text{HCl}) \cdot V(\text{HCl})}{V(\text{NaOH})}$ $c(\text{NaOH}) = \frac{0,0998 \text{ mol dm}^{-3} \cdot 22,80 \text{ cm}^3}{25,00 \text{ cm}^3} = 0,0910 \text{ mol dm}^{-3}$	1 BOD
8.1.	izolirani	1 BOD
8.2.	endotermna kemijska promjena	1 BOD

8.3.		1 BOD
9.1.	H ₂	1 BOD
9.2.	Al ³⁺ + 3 OH ⁻ → Al(OH) ₃	1 BOD
9.3.	[Al(OH) ₄] ⁻ , [Al(OH) ₄ (H ₂ O) ₂] ⁻	1 BOD
9.4.	SO ₄ ²⁻ (aq) + Ba ²⁺ (aq) → BaSO ₄ (s)	1 BOD
10.1.	 $\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{H}_2\text{SO}_4} \text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 + \text{H}_2\text{O}$	1 BOD
10.2.	eliminacija	1 BOD
10.3.	halogenalkanima ili alkil-halogenidima	1 BOD
10.4.	eter	1 BOD
11.1.	magnezij, Mg	1 BOD
11.2.	Cu(s) Cu ²⁺ (aq) Ag ⁺ (aq) Ag(s)	1 BOD
11.3.	Cu ²⁺ + Fe → Cu + Fe ²⁺	1 BOD
11.4.	U vodenoj otopini srebrove soli.	1 BOD

12.1.	tvar A	1 BOD
12.2.	$\bar{v} = -\frac{1}{2} \frac{\Delta c(B)}{\Delta t}$	1 BOD
12.3.	$\bar{v}(C) = 0,075 \pm 0,025 \text{ mmol dm}^{-3} \text{ min}^{-1}$ $\bar{v}(C) = \frac{c_2 - c_1}{t_2 - t_1} = \frac{(5,00 - 4,7) \text{ mmol dm}^{-3}}{8 \text{ min} - 4 \text{ min}} = 0,075 \text{ mmol dm}^{-3} \text{ min}^{-1}$	1 BOD
12.4.	Vrijednost reakcijske entalpije ostat će jednaka.	1 BOD